

Google Cloud

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Google Cloud Skills Boost for Partners

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Integrate Vertex AI Search and Conversation into Voice and Chat Apps

Course · 5 hours · 60% complete

Build Vertex AI Search Apps using AI Applications

Enable informed decision making with a conversational agent that uses generators and data stores

Conversational Agents with Generative Fallbacks

Create a chat and voice FAQ bot with Conversational Agent Playbooks and Data Stores

Search and Conversational AI in Vertex AI and Conversational Agents: Challenge Lab

Your Next Steps

Course > Integrate Vertex AI Search and Conversation into Voice and Chat Apps >

Quick tip: Review the prerequisites before you run the lab

End Lab 01:07:10

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud Console

Username

student-02-2888b80c7d70i

Password

FYrFa8UoowGq

GCP Project ID

qwiklabs-gcp-02-1833d8fi

Create a chat and voice FAQ bot with Conversational Agent Playbooks and Data Stores

Lab 1 hour 30 minutes No cost Intermediate

★★★★☆ This lab may incorporate AI tools to support your learning.

Lab instructions and tasks 100/100

Overview

Setup and requirements

Task 1. Enable APIs

Task 2. Create a Conversational Agent Playbook and a data store

Task 3. Test your virtual agent

Task 4. Enable voice calls

Task 5. Enable chat widget

Task 6. Conversation analytics

Congratulations!

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## What you'll build

In this lab, you'll use [Vertex AI Agent Builder data stores](#) to build, configure, and deploy a Conversational Agent that can assist customers who have questions about products and devices in the [Google Store](#), including phones, watches, laptops, smart home devices, and other consumer devices. Additionally, you'll connect you agent to a Conversational Phone Gateway to enable voice calls and a Chat Widget to enable customized chat solutions within your website.

## What are data stores?

Data stores are used by Conversational Agents to find answers for end-user's questions from your data. Data stores are a collection of websites and documents, each of which reference your data.

With a data store, you can provide a website URL, structured data, or unstructured data, then the data store parses and indexes your content. Then from Conversational Agents you can reference data stores from Playbook using tools and from Flows using data store handlers or webhooks. Your customers and end users can then have conversations with the *Conversational Agent* and ask questions about the content.

An illustration showing two stylized human figures standing next to a large laptop. One figure is pointing at the screen, which displays a magnifying glass icon over a document. The other figure is holding a tablet. In the background, there is a small building and a ladder, suggesting a construction or development process.

## Objectives

In this lab, you will learn how to perform the following tasks:

- Use Conversational Agents to create a Playbook
- Create a data store for your Playbook and add documents to it
- Test your agent and simulate customer questions
- View conversation history and analytics for your agent

## Setup and requirements

In this task, you use Qwiklabs and perform initialization steps for your lab.

For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Make sure you signed into Qwiklabs using an **incognito window**.
2. Note the lab's access time (for example, **02:00:00**) and make sure you can finish in that time block.

There is no pause feature. You can restart if needed, but you have to start at the beginning.

3. When ready, click

**START LAB**

4. Note your lab credentials. You will use them to sign in to the Google Cloud

Open Google Console

**Caution:** When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

**Username**  
google2876526\_student@qwiklabs.n

**Password**  
TG959yrKDX

qwiklabs-gcp-0855e773352d3560

New to labs? [View our introductory video!](#)

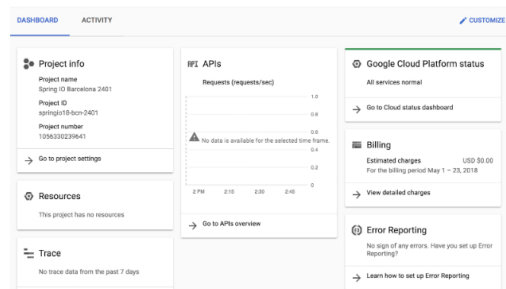
5. Click **Open Google Console**.
6. Click **Use another account** and copy/paste credentials for **this** lab into the prompts.

If you use other credentials, you'll get errors or **incur charges**.

7. Accept the terms and skip the recovery resource page.

Do not click **End Lab** unless you are finished with the lab or want to restart it. This clears your work and removes the project.

## Google Cloud console

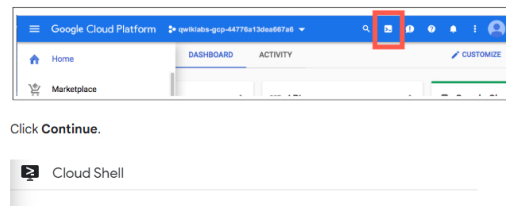


- Click **Select a project**, highlight your Google Cloud project ID, and click **OPEN** to select your project.

## Activate Cloud Shell

command-line access to your Google Cloud resources.

In the Cloud Console, in the top right toolbar, click the **Activate Cloud Shell** button.

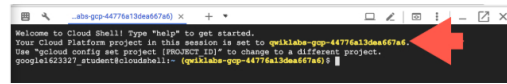


Click **Continue**.

Google Cloud Shell provides you with command-line access to your cloud resources directly from your browser. You can easily manage your projects and resources without having to install the Google Cloud SDK or other tools on your system. [Learn more](#).

Continue

connected, you are already authenticated, and the project is set to your *PROJECT\_ID*. For example:



```
Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to qwiklabs-gcp-44776a13dea667a6.
Use "gcloud config set project [PROJECT_ID]" to change to a different project.
google162327_student@cloudshell:~ (qwiklabs-gcp-44776a13dea667a6) $
```

gcloud is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

You can list the active account name with this command:

gcloud auth list

(Output)

```
Credentialed accounts:
- <myaccount>@<mydomain>.com (active)
```

```
Credentialed accounts:
- google162327_student@qwiklabs.net
```

You can list the project ID with this command:

gcloud config list project

(Output)

```
[core]
project = <project_ID>
```

(Example output)

```
[core]
project = qwiklabs-gcp-44776a13dea667a6
```

For full documentation of gcloud, see the [gcloud command-line tool overview](#).

## Task 1. Enable APIs

In this task, you enable the Conversational Agents (prev. known as Dialogflow CX) and Vertex AI Agent Builder APIs in your Google Cloud project.

To enable the Conversational Agents APIs, follow these steps:

1. In your browser, navigate to the [Dialogflow API Service Details page](#).
2. If the **Enable** button is visible, then click the **Enable** button to enable the Dialogflow API in your Google Cloud project. If the **Enable** button is not visible, then the API is already enabled in your project.

To enable the Vertex AI Agent Builder APIs, follow these steps:

1. In the Google Cloud console, navigate to the [Vertex AI Agent Builder console](#).

Click **Check my progress** to verify the objectives.



Enable APIs

Check my progress

Assessment Completed!

## Task 2. Create a Conversational

Now, you'll create a new Conversational Agent Playbook and configure it with a data store. The purpose of the agent that you'll build is to assist customers who have questions about products in the [Google Store](#).

1. To create a new **Conversational Agent**, in [Vertex AI Agent Builder Apps Console](#) choose **Conversational agent** as the app type and click **Create**.
2. Click **Create agent**.
3. Select **Build your own** on Get started with Conversational Agents popup.
4. On **Create agent** page, enter a **Display name** as **Google Store**.
5. Select location as **global (Global serving, data-at-rest in US)**.
6. Ensure the default **Conversational start** type is selected as **Playbook**, and then click on **Create** button.
7. Click **+ Data store** under **Available tools** in Default Generative Playbook page.

9. Click **Create new data store** in the pop-up window, and you will be redirected to the Agent Builder page.

10. Select **Cloud Storage** as the data source for your data store.

11. Select **Unstructured documents** as the type of data you are importing.

**Note:** Currently, in the bucket there are html documents related to various Google

13. Click **Continue**.

14. Specify a **Data store name** as Google Store.

15. Click **Create** to create the data store.

16. In the list of data stores, select the newly created data store named Google Store.

17. Click on the **Activity** tab to see the progress of the data import.

**Note:** It can take up to 15 minutes for your documents to be available and ready for

Next, it can take up to 10 minutes for your documents to be processed and ready for use by your agent.

18. Once the agent is created, navigate back to the **Conversational Agents Tools** console and refresh the page.

19. Enter the `Google_store_tool` in the **Tool name** field and choose **Data store** as

Create Tool [New](#)

Using tools, you can connect playbooks to external systems. These options can augment the knowledge of playbooks and empower them to execute complex tasks efficiently. [Learn more](#)

Tool name:

Data store:

Other tools can be used by a generative agent to answer to end-user's questions from your data stores. Note: Conversation history is used as context during tool invocation. [Learn more](#)

Description

Provide a description of the tool. This description is provided to the model as context informing how the tool is used.

Description:

Data stores

Add data stores

Filter Search data stores

| Name         | Create Time | Type                   |
|--------------|-------------|------------------------|
| Google Store | Mar 6, 2025 | Unstructured documents |

20. Under **Data Stores**, click on **Add Data Stores**, check the checkbox for **Google Store**, then click **Confirm** and finally, click **Save**.

Add data stores

You can only select up to one collection of FAQ, website, and unstructured data store type.

[Create new data store](#)

1 data store selected.

Filter Search data stores

|                                     |              |             |                        |
|-------------------------------------|--------------|-------------|------------------------|
| <input checked="" type="checkbox"/> | Google Store | Mar 6, 2025 | Unstructured documents |
|-------------------------------------|--------------|-------------|------------------------|

[Confirm](#) [Cancel](#)

21. Navigate to the **Playbooks**, then click on **Default Generative Playbook**. Under **Available tools** choose the tool called `Google_store_tool` and click **Save**.

Conversational Agents [Project](#) [Agent](#) [Language](#)

[Default Generative Playbook](#) [Version history](#) [Save](#)

Basics Examples Settings

Playbook name:

A playbook is the basic building block of a Vertex AI Conversation app. Each playbook is defined to handle specific tasks. [Learn more](#)

Goal

Goal:

High level description of the goal the playbook intends to accomplish. [Learn more](#)

Instructions

Instructions:

Available tools

This playbook can use selected tools to generate responses. You can also call other tools, including 3P, directly in step questions using data store content.

+ Data store [Manage all tools](#)

☒ Google\_store\_tool [View info](#)

☐ code interpreter [View info](#)

Congratulations! You have finished building your knowledge-powered conversational agent that's ready to help your customers, so take a moment to celebrate!

But there's still more work to do to make the bot accessible to your users. In the next section, you'll test your virtual agent and see how good it is at answering user questions about various products in the Google Store.

Click **Check my progress** to verify the objectives.

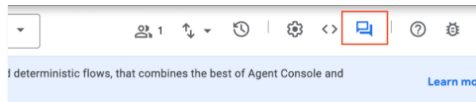
Create a new app

[Check my progress](#)

## Task 3. Test your virtual agent

Testing is an important part of working with conversational agents to find bugs, identify limitations, and simulate a customer experience. You can start an interactive session with your chatbot to see how it responds to various questions that a customer might ask.

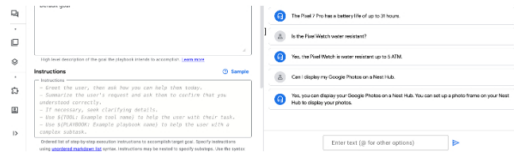
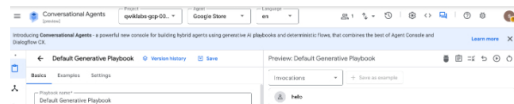
1. In the Conversational Agents console within your agent, click **Toggle Simulator** icon to open the Simulator.



3. Ask the agent some questions about various products such as:

- How long does the battery in the Pixel 7 Pro last?
- Is the Pixel Watch water resistant?
- Can I display my Google Photos on a Nest Hub.
- Ask other questions about doorbells, thermostats, smart lighting, or other devices in the Google Store.

**Note:** If your agent is giving you responses from the default negative intent (e.g., "I'm sorry, I didn't get that." or "Can you rephrase your question?"), then be aware that it can take up to 4 hours for the agent to be ready for use while your domains, URLs, or documents are being indexed. Your documents should be indexed for the sake of this lab within 15 minutes.



Great work! Notice that your virtual agent knows quite a bit about different products in the Google Store since it ingested information from the root domain that you specified earlier. Refer to the documentation for the [agent simulator](#) to learn more about testing your agent.

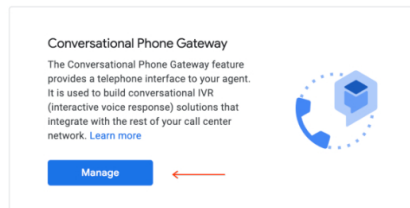
## Task 4. Enable voice calls

Now that you've tested your agent and are happy with its current level of functionality, you can add a phone gateway to your bot, which will make use of the [Speech-to-Text](#)

1. In the Conversational Agents console within your agent, click on **Integrations** tab from the sidebar.
2. On the **Integrations** page, click **Manage** within the **Conversational Phone Gateway** section.

### Integrations

#### Conversational Phone Gateway



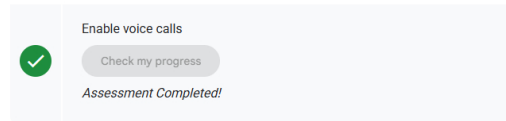
3. On the **Phone numbers** page, click **Create new** to start the process of creating a

4. Select a country code and click **Request** to continue to the next step.
5. Select one of the phone numbers that appears, input a display name, then click **Save**.
6. Call your agent using the selected phone number from your own phone and ask it some questions!

- Do the Pixel 7 Pro and Pixel 7 both have face unlock?
- What coverage does Preferred Care provide for a Pixelbook Go laptop?
- Do the Pixel Buds Pro have active noise cancellation?
- Ask other questions about doorbells, thermostats, smart lighting, or other devices in the Google Store.

Congratulations, you gave your virtual agent its own phone number and voice! Refer to the documentation for [Conversational Agents Integrations](#) for more information on other available voice integrations.

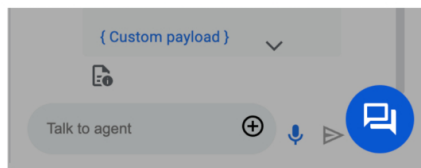
Next, you'll integrate a chat messenger for your virtual agent into an external website.



## Task 5. Enable chat widget

Now that your bot has a phone gateway for voice interactions, let's embed a chat widget on a website so customers can chat with it in addition to making a phone call to speak with it.

1. Go back to the **Integrations** tab in the Conversational Agents console within your agent.
2. Click **Connect** under **Conversational Messenger** in the **Text Based** section of
3. Click **Enable the unauthenticated API** in the dialog.
4. Click **Try it now** and then click the Conversational Messenger icon on the bottom-right of your window to expand the Conversational Messenger application.

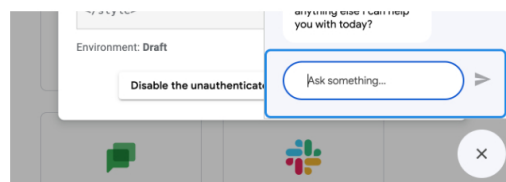
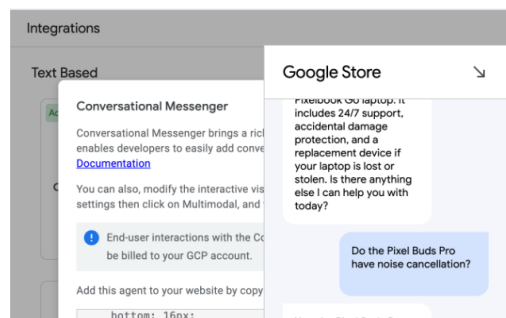


5. Test and chat with your virtual agent! Here are some sample questions to ask:

- Do the Pixel 7 Pro and Pixel 7 both have face unlock?
- How does Preferred Care relate to a Pixelbook Go laptop?

- Ask other questions about doorbells, thermostats, smart lighting, or other devices in the Google Store.

Now your virtual agent can now handle questions and answers from your customers via chat or voice, whichever they prefer! For more information on other available chat integrations, refer to the documentation for [Conversational Agents Integrations](#).



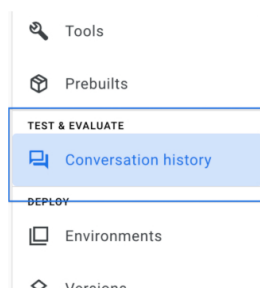
## Task 6. Conversation analytics

Other than testing many different paths that customers might take, how else can you

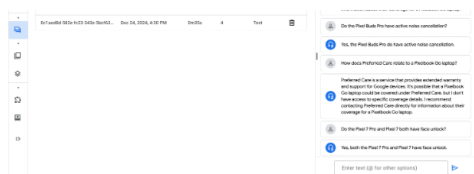
- 

- Logging settings**
- ☐ **Enable Cloud Logging**  
Export user queries and debugging information to [Cloud Logging](#).
- ☒ **Indicate Conversation History**  
Indicates whether you would like [Logging](#) to collect and store embedded end-user queries for quality improvement.  
Note some features, such as [Experimentation](#), [export to BigQuery](#) and [ML model](#) improvement, require this setting to be enabled, as they rely on traffic logs. Production needs to be configured by the developer through [ML or experiment submodules](#).

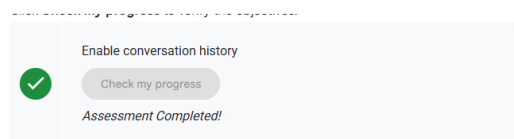
- click the **Conversation history** tab.



- 



- This data helps you assess how your agent is being used in production and can be used to determine which websites and documents you might want to add to your knowledge base to improve your agent and customer experience.



By using data stores in Vertex AI Agent Builder and Conversational Agents, you've successfully created a virtual agent, added data sources, and deployed a voice-enabled



...necessarily created a virtual agent, added data sources, and deployed a voice-enabled and chat-enabled agent to help your customers!

Your virtual agent can answer hundreds of different questions about products in the Google Store, and you didn't have to go through the manual process of creating a large number of intents, training phrases, response messages, etc.

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functionality available related to [Vertex AI Agent builder](#) and [Conversational Agents](#).

## Learn more

Continue learning about conversational AI and generative AI with these guides and resources:

- [Documentation for Conversational Agents](#)
- [Overview of Agent Builder](#)
- [Documentation for data stores](#)
- [Documentation for Agent Builder](#)
- [Generative AI on Google Cloud](#)

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**Lab Last Tested March 12, 2025**

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